COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2021 – Paper 7

9 Further Human–Computer Interaction (afb21)

Current directions - augmented reality

(*a*) Augmented reality displays such as the Microsoft Hololens allow elements of the display to be registered against objects in the physical world. Briefly explain how such a display could be used for an Internet of Things application, setting different temperatures for multiple rooms in a home heating system. [1 mark]

Answer: You could look at the heater and see a temperature slider in the augmentation, using a gesture to move it up and down. You have to go to each room and repeat for that room.

Design of visual displays: visual metaphor

(*b*) Explain how this compares to the use of visual metaphor on a conventional display for the same application, taking account of the correspondence between graphical resources and their representational purpose. [3 marks]

Answer: In a visual metaphor, the slider appears on the screen, with upward direction corresponding to increased temperature as a scale value. The metaphor could use a perspective rendering as a pictorial representation of going into different rooms. However, rather than simulating the experience, it would be better to lay out the screen as a map of the house, perhaps with a slider rendered as a conventional icon within the framed region indicated by the walls of each room.

Variables of the display plane and modes of correspondence

(*c*) Describe an alternative approach to visual representation that could be more efficient for setting desired temperature in a large number of rooms, for example in a multi-storey office building. Your description should mention three different categories of graphic resource, explaining the design principles for each of them in relation to this application. [8 marks]

Answer: For a large number of rooms, the direct manipulation metaphor will be inefficient. It would be necessary to design some kind of abstraction so that the user can define categories of room that should have the same temperature. One region of the screen could be used for a map overview of the building, with colour codes assigned to each room to show which category the room is in. The colour scale could be ordered by hue, using the conventional temperature scale where blue is cold and red is hot. The temperature settings for each room category would be defined in a separate region of the screen, separated from the map by some white space. Multiple sliders could be ordered into a list, with a colour patch corresponding to each slider by alignment, or by topological connection drawing a line between them.

Designing complex systems

(*d*) Choose three cognitive dimensions that are especially relevant to this problem, and explain how these dimensions would be different if the temperature settings were being adjusted using a speech interface such as Amazon Alexa or Google Assistant. [6 marks]

Answer: Viscosity is relevant to changing the temperature. The direct manipulation interface is more viscous, because each room has to be adjusted separately. The use of room categories is an abstraction, which reduces viscosity, but could involve premature commitment because you have to define the categories before setting the actual temperatures. When using a voice assistant, the category would have to be given a name, which increases the need for abstraction.

It also adds hidden dependencies, because you can't see which rooms are in the category, meaning that an adjustment to the temperature could have unexpected effects in other rooms that you forgot. Voice interaction could be a lot more longwinded (diffuse) in a large building, where you have to describe the location of each room rather than just pointing to it.

Qualitative research - grounded theory

(*e*) How could an ethnographic approach be used to develop an alternative theory of correspondence relevant to this problem? [2 marks]

Answer: You could use ethnographic methods to get a better understanding of how people think about the structure of their house, talk about temperature in relation to that structure, and assign responsibility to the person who defines temperature policies. Transcribing field recordings and using grounded theory analysis could help to find new kinds of correspondence in these situations.
